# KAPU measurements

On a knowledge file the model had not seen (PixelLab documentation, 206,899 tokens), an agent using KAPU read 4,104 tokens per request, an agent searching one Markdown file 8,280, and one searching the same text as a PDF 8,207 (median, the same prompt for all three). Correct answers: KAPU 12, Markdown 12, PDF 11 of 12, by three blind judges.

Latest measurements 2026-10-28 · tools 0.5.0, specification v1.3, format 1.3 · Windows 11, release build · every number comes from bench/results/ in the repository

| KAPU | One .md file + search | PDF + search | Whole file in the context |
|---|---|---|---|
| 4,104 | 8,280 | 8,207 | 206,899 |
| tokens per request (median) | tokens per request (median) | tokens per request (median) | tokens on every request |

## Plan v3: a contextual model in plain Rust (0.5.0)

Measured 2026-10-28 with tools 0.5.0, specification v1.3, format 1.3, on the seven question sets and the dev/test split of plan v2 (577 test questions), the test part reported. The static model of 0.3 limited the hybrid search, so plan v3 measured other embedding models and built the winner, multilingual-e5-small, into KAPU: its own Rust code runs the model with exact integer products, so a vector has the same bits on every machine, and the file stores every vector at 4 bits a component.

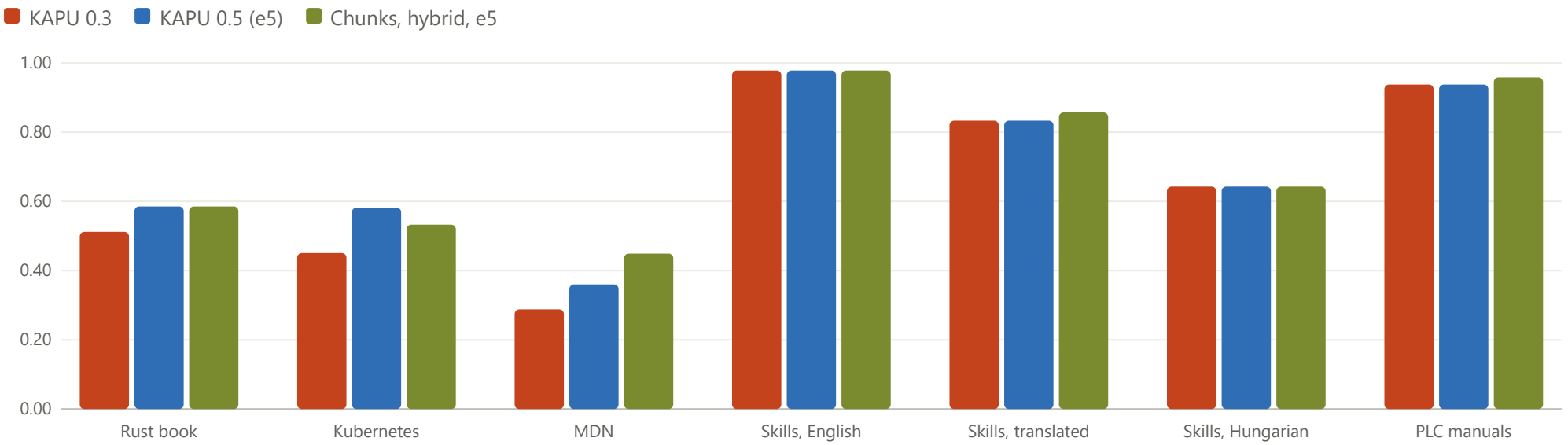
## The model swap in KAPU's own search

Difference to the static model of 0.3 with its default ranking, test part, all sets pooled, 95% paired bootstrap interval. Worst set: the largest loss of one set in either metric; p50 ratio: the slowest set's median search time against 0.3. The swap does not pass the M0 rule of 0.3.0 (worst set at most -0.01, at most 1.5 times slower); it was accepted for 0.5.0 with these costs.

| Model | Bits | Ranking | Δ room recall@10 | Δ room MRR@10 | Worst set | p50 ratio |
|---|---|---|---|---|---|---|
| multilingual-e5-small | 8 | Default ranking | +0.062 [+0.035, +0.090] | +0.027 [+0.005, +0.049] | −0.065 | 14.9× |
| multilingual-e5-small | 4 | Default ranking | +0.062 [+0.035, +0.090] | +0.028 [+0.006, +0.052] | −0.056 | 14.9× |
| multilingual-e5-small | 4 | Passage BM25 | +0.069 [+0.038, +0.101] | +0.038 [+0.014, +0.062] | −0.070 | 14.6× |
| multilingual-e5-small | 4 | Passage BM25 and room fusion | +0.075 [+0.045, +0.106] | +0.038 [+0.014, +0.063] | −0.085 | 14.5× |

## Per set: 0.3, 0.5 and the chunked hybrid with the same model

Room-level recall@10 on the test part, default ranking, 4 bits. Search time: the median of kapu bench, with the query's vector built for AVX-512, and with the baseline (SSE2) build of the release.




| Question set | Questions | recall@10, 0.3 | recall@10, 0.5 | MRR@10, 0.3 | MRR@10, 0.5 | Search, 0.3 | Search, 0.5 | 0.5, SSE2 |
|---|---|---|---|---|---|---|---|---|
| Rust book | 41 | 0.512 | 0.585 | 0.252 | 0.263 | 0.8 ms | 12.6 ms | 35.0 ms |
| Kubernetes | 122 | 0.451 | 0.582 | 0.253 | 0.271 | 27.1 ms | 25.7 ms | 48.4 ms |
| MDN | 236 | 0.288 | 0.360 | 0.135 | 0.195 | 408.5 ms | 155.9 ms | 320.4 ms |
| Skills, English | 46 | 0.978 | 0.978 | 0.853 | 0.868 | 10.1 ms | 17.2 ms | 43.1 ms |
| Skills, translated | 42 | 0.833 | 0.833 | 0.759 | 0.770 | 12.7 ms | 17.6 ms | — |
| Skills, Hungarian | 42 | 0.643 | 0.643 | 0.492 | 0.436 | 10.7 ms | 18.9 ms | — |
| PLC manuals | 48 | 0.938 | 0.938 | 0.734 | 0.751 | 29.7 ms | 30.1 ms | — |

## What it costs: file size and build time

The first build with the model on 12 threads (AVX-512 build). A rebuild takes the computed windows from a log in the cache folder and computes only new text.

| File | Size, 0.3 | Size, 0.5 | Growth | With 8 bits | First build |
|---|---|---|---|---|---|
| Rust book | 2.4 MB | 2.6 MB | +9% | +20% | 71 s |
| Kubernetes | 28.1 MB | 31.4 MB | +12% | +25% | 922 s |
| MDN | 139.0 MB | 160.7 MB | +16% | +34% | 2,986 s |
| Skills, English | 5.9 MB | 6.5 MB | +10% | +21% | 124 s |
| PLC manuals | 12.3 MB | 13.4 MB | +9% | +20% | 321 s |

## How to store a vector (simulated on the chunks)

KAPU BM25 fused with the chunk vectors, difference to float vectors. Centred: the mean of all vectors taken off first; residual: the code of the vector minus the mean, scored with the original cosine. The sign bits of 0.3 break down with this model, 4 bits cost nothing measurable in KAPU itself.

| Storage | Bytes per window | Δ room recall@10 | Δ room MRR@10 |
|---|---|---|---|
| float, centred | 1,536 | −0.021 [−0.042, +0.000] | −0.002 [−0.015, +0.011] |
| sign bits, then 8 bits | 432 | **−0.121 [−0.154, −0.088]** | **−0.066 [−0.089, −0.044]** |
| 8 bits | 384 | +0.002 [+0.000, +0.005] | **+0.003 [+0.001, +0.006]** |
| sign bits, then 4 bits | 240 | **−0.127 [−0.159, −0.095]** | **−0.074 [−0.096, −0.051]** |
| centred sign bits, then 4 bits | 240 | −0.012 [−0.033, +0.009] | −0.001 [−0.014, +0.011] |
| 4 bits, residual | 194 | −0.009 [−0.023, +0.005] | −0.002 [−0.010, +0.006] |
| 4 bits | 192 | −0.012 [−0.028, +0.003] | −0.001 [−0.010, +0.007] |
| 4 bits, centred | 192 | −0.017 [−0.038, +0.003] | −0.004 [−0.016, +0.009] |
| 3 steps, centred | 144 | −0.019 [−0.040, +0.002] | +0.002 [−0.011, +0.015] |
| -1/0/+1, residual | 98 | **−0.035 [−0.055, −0.014]** | **−0.024 [−0.041, −0.007]** |
| -1/0/+1, centred | 96 | −0.021 [−0.043, +0.002] | −0.005 [−0.018, +0.009] |
| sign bits, residual | 50 | **−0.080 [−0.109, −0.052]** | **−0.063 [−0.085, −0.041]** |
| sign bits (as in 0.3) | 48 | **−0.203 [−0.239, −0.168]** | **−0.178 [−0.208, −0.149]** |
| sign bits, centred | 48 | **−0.033 [−0.055, −0.010]** | −0.010 [−0.025, +0.005] |

### The candidates on the same chunks (V1)

Three models on the 512-token chunks, in Python, difference to the static model of 0.3: the vectors alone, the usual chunked hybrid, and KAPU's BM25 ranking fused with the model's vectors at room level.

| Model | Ranking | Δ room recall@10 | Δ room MRR@10 |
|---|---|---|---|
| potion-retrieval-32M | Vectors alone | +0.021 [−0.017, +0.059] | +0.016 [−0.010, +0.042] |
| potion-retrieval-32M | Chunked hybrid | −0.007 [−0.033, +0.019] | +0.002 [−0.019, +0.023] |
| potion-retrieval-32M | KAPU BM25 + vectors | +0.003 [−0.026, +0.033] | +0.009 [−0.013, +0.032] |
| potion-retrieval-32M | KAPU passage BM25 + vectors | +0.009 [−0.021, +0.038] | +0.022 [−0.000, +0.044] |
| multilingual-e5-small-qint8 | Vectors alone | **+0.123 [+0.083, +0.163]** | **+0.072 [+0.042, +0.102]** |
| multilingual-e5-small-qint8 | Chunked hybrid | **+0.064 [+0.036, +0.092]** | **+0.039 [+0.016, +0.062]** |
| multilingual-e5-small-qint8 | KAPU BM25 + vectors | **+0.071 [+0.040, +0.102]** | **+0.039 [+0.016, +0.063]** |
| multilingual-e5-small-qint8 | KAPU passage BM25 + vectors | **+0.075 [+0.045, +0.104]** | **+0.053 [+0.030, +0.076]** |

- The model is the lever: with multilingual-e5-small, room recall@10 improves by +0.062 with the default ranking and by +0.075 with passage BM25 and room fusion, both intervals above zero. Kubernetes 0.451 to 0.582, MDN 0.288 to 0.360.
- It costs: Hungarian questions on English skills lose MRR (0.492 to 0.436), a small file is searched 12.6 ms instead of 0.8 ms (the query's vector; 35.0 ms with the baseline build), and the files grow by 9-16%. A large file is faster (408.5 ms to 155.9 ms), because nothing is rescored when it is read.
- With the same model, the usual chunked hybrid is 0.585 on the Rust book, 0.533 on Kubernetes and 0.449 on MDN: level with KAPU 0.5, behind it and ahead of it. KAPU's advantage stays the section hit with its token price, in one file with no server.
- The encoder is KAPU's own Rust code, with no new dependency and nothing to download at run time: integer products, fixed-order float sums and its own exp and erf, so the SSE2, AVX2 and AVX-512 builds write the same bytes. It agrees with ONNX Runtime to a cosine of 0.998-1.000; one 512-token window takes 0.45 s on one thread (AVX-512), 1.3 s with the baseline build.

## Plan v2, phases 1 to 3: an ablation framework, one fix that holds, honest baselines

Measured 2026-10-27 with tools 0.3.0. Every ranking change of plan v2 is a switch that is off by default; off, the ranking is bit for bit that of 0.2.1. The questions of each set are split once by a

hash of the question into a dev part (40%, 356 questions in all) and a test part (60%, 577): the variant of each fix is chosen on the dev part, and only the test part is reported. A fix is kept when on the test part room recall@10 or room MRR@10 improves with a 95% paired bootstrap interval above zero (all sets pooled), no set loses more than 0.01 and search does not slow down by more than 1.5x.

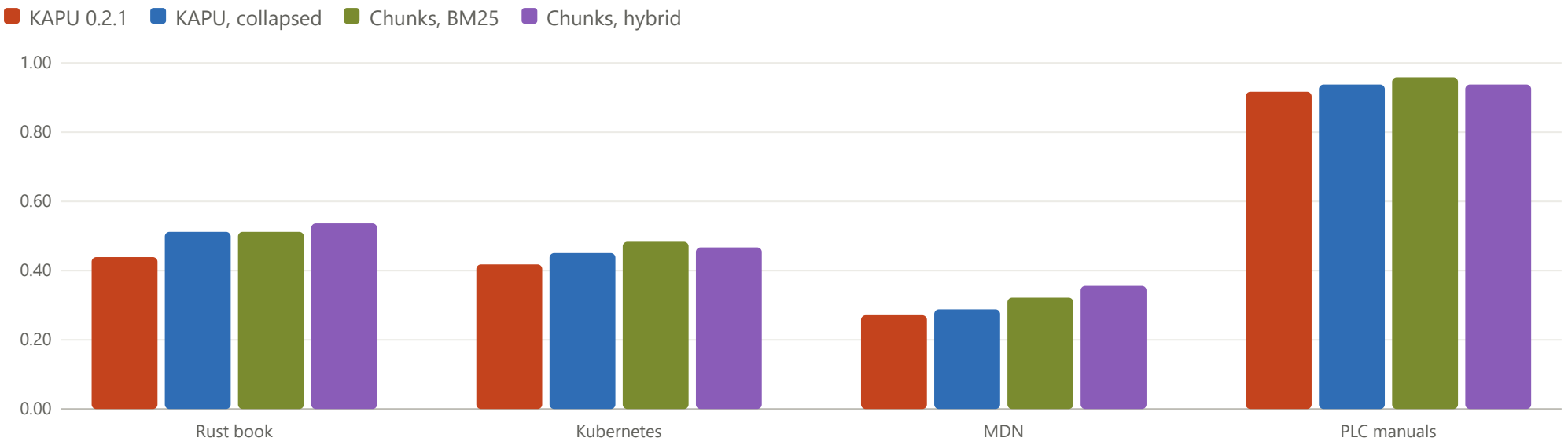
## Which fixes pass

Difference to the base ranking on the test part, all sets pooled, with the 95% bootstrap interval. Green: the interval is above zero; red: below. Worst set: the largest loss of one set in either metric.

| Fix | Variant chosen on dev | Δ room recall@10 | Δ room MRR@10 | Worst set | Kept |
|---|---|---|---|---|---|
| Stemming: the word only, English or Hungarian stem (M0 axis) | stem-en | +0.002 [−0.012, +0.016] | −0.005 [−0.014, +0.004] | −0.021 | no |
| J2a: separate average length for rooms and sections | avgdl-split | +0.021 [+0.005, +0.036] | +0.016 [+0.003, +0.028] | −0.022 | no |
| J2b: a section scores a share of its room | parent-0.1 | −0.004 [−0.019, +0.012] | −0.009 [−0.018, −0.000] | −0.030 | no |
| J1: one hit per room, the other matching sections named | collapse-1 | +0.022 [+0.010, +0.035] | +0.011 [+0.008, +0.014] | +0.000 | yes |
| J4a: the first 50 BM25 hits always get a vector score | rescore-bm25-50 | +0.002 [−0.005, +0.010] | +0.001 [−0.002, +0.005] | −0.021 | no |
| J10: sections under 40 tokens only through their room | min-door-40 | −0.002 [−0.009, +0.004] | +0.001 [−0.001, +0.004] | −0.009 | no |
| J4c: weights by query (code words, unknown words) | adaptive-1.5 | +0.002 [−0.005, +0.009] | +0.000 [−0.003, +0.003] | −0.021 | no |
| J4b: weighted RRF (BM25 1-3, vectors 0.5-1, k 20 or 60) | rrf-b1-v0.7-k60 | +0.009 [−0.005, +0.024] | +0.006 [+0.001, +0.012] | −0.048 | no |
| J4d: min-max scores instead of ranks | convex-b1-v1 | +0.002 [−0.017, +0.021] | +0.004 [−0.008, +0.016] | −0.024 | no |

## Retrieval against the chunk baselines (test part)

Room-level: a hit is any door of the expected page. The chunk baselines cut every page into 512-token chunks and count a page once (deduplicated), which is the fair comparison with KAPU's room level. Collapsed: one hit per room (J1), the new default.




| Question set | Questions | Ranking | recall@1 | Recall@10 | recall@100 | MRR@10 | nDCG@10 |
|---|---|---|---|---|---|---|---|
| Rust book | 41 | KAPU 0.2.1 | 0.146 | 0.439 | 0.902 | 0.220 | 0.271 |
| Rust book | 41 | KAPU, collapsed | 0.146 | 0.512 | 0.902 | 0.252 | 0.314 |
| Rust book | 41 | Chunks, BM25 | 0.146 | 0.512 | 0.902 | 0.255 | 0.315 |
| Rust book | 41 | Chunks, vectors | 0.122 | 0.439 | 0.854 | 0.223 | 0.274 |
| Rust book | 41 | Chunks, hybrid | 0.146 | 0.537 | 0.829 | 0.249 | 0.317 |
| Kubernetes | 122 | KAPU 0.2.1 | 0.139 | 0.418 | 0.713 | 0.238 | 0.282 |
| Kubernetes | 122 | KAPU, collapsed | 0.139 | 0.451 | 0.713 | 0.253 | 0.302 |
| Kubernetes | 122 | Chunks, BM25 | 0.164 | 0.484 | 0.795 | 0.242 | 0.299 |
| Kubernetes | 122 | Chunks, vectors | 0.066 | 0.328 | 0.639 | 0.145 | 0.189 |
| Kubernetes | 122 | Chunks, hybrid | 0.164 | 0.467 | 0.738 | 0.250 | 0.301 |
| MDN | 236 | KAPU 0.2.1 | 0.076 | 0.271 | 0.525 | 0.127 | 0.161 |
| MDN | 236 | KAPU, collapsed | 0.076 | 0.288 | 0.525 | 0.135 | 0.171 |

| Question set | Questions | Ranking | recall@1 | Recall@10 | recall@100 | MRR@10 | nDCG@10 |
|---|---|---|---|---|---|---|---|
| MDN | 236 | Chunks, BM25 | 0.089 | 0.322 | 0.602 | 0.156 | 0.196 |
| MDN | 236 | Chunks, vectors | 0.064 | 0.203 | 0.517 | 0.095 | 0.120 |
| MDN | 236 | Chunks, hybrid | 0.064 | 0.356 | 0.631 | 0.145 | 0.195 |
| Skills, English | 46 | KAPU 0.2.1 | 0.783 | 0.978 | 1.000 | 0.846 | 0.878 |
| Skills, English | 46 | KAPU, collapsed | 0.783 | 0.978 | 1.000 | 0.853 | 0.884 |
| Skills, translated | 42 | KAPU 0.2.1 | 0.714 | 0.833 | 0.833 | 0.756 | 0.775 |
| Skills, translated | 42 | KAPU, collapsed | 0.714 | 0.833 | 0.833 | 0.759 | 0.777 |
| Skills, Hungarian | 42 | KAPU 0.2.1 | 0.405 | 0.619 | 0.762 | 0.487 | 0.520 |
| Skills, Hungarian | 42 | KAPU, collapsed | 0.405 | 0.643 | 0.762 | 0.492 | 0.529 |
| PLC manuals | 48 | KAPU 0.2.1 | 0.625 | 0.917 | 0.979 | 0.720 | 0.767 |
| PLC manuals | 48 | KAPU, collapsed | 0.625 | 0.938 | 0.979 | 0.734 | 0.783 |
| PLC manuals | 48 | Chunks, BM25 | 0.625 | 0.958 | 0.979 | 0.741 | 0.794 |
| PLC manuals | 48 | Chunks, vectors | 0.458 | 0.792 | 0.896 | 0.571 | 0.625 |
| PLC manuals | 48 | Chunks, hybrid | 0.583 | 0.938 | 0.979 | 0.686 | 0.746 |

## Why a question is missed (M3)

Every test question whose page is not in the hybrid top 10, by the first class that fits: duplication (the page would be in the top 10 with one hit per room), fusion (one ranking had it in its top 10), rescoring (a good BM25 hit without a vector score), ranking (in the top 100), vocabulary (nowhere). Merge: the page is in the top 10 but the expected section is not.

| Question set | Ranking | Missed | Duplication | Fusion | Rescoring | Ranking | Vocabulary | Merge |
|---|---|---|---|---|---|---|---|---|
| Rust book | KAPU 0.2.1 | 23 | 3 | 5 | 1 | 10 | 4 | 1 |
| Rust book | KAPU, collapsed | 20 | 0 | 5 | 1 | 10 | 4 | 8 |
| Kubernetes | KAPU 0.2.1 | 71 | 4 | 11 | 6 | 15 | 35 | 9 |
| Kubernetes | KAPU, collapsed | 67 | 0 | 11 | 6 | 15 | 35 | 26 |
| MDN | KAPU 0.2.1 | 172 | 4 | 17 | 15 | 24 | 112 | 8 |
| MDN | KAPU, collapsed | 168 | 0 | 17 | 15 | 24 | 112 | 23 |
| Skills, English | KAPU 0.2.1 | 1 | 0 | 0 | 0 | 1 | 0 | 2 |
| Skills, English | KAPU, collapsed | 1 | 0 | 0 | 0 | 1 | 0 | 15 |
| Skills, translated | KAPU 0.2.1 | 7 | 0 | 0 | 0 | 0 | 7 | 1 |
| Skills, translated | KAPU, collapsed | 7 | 0 | 0 | 0 | 0 | 7 | 7 |
| Skills, Hungarian | KAPU 0.2.1 | 16 | 1 | 1 | 0 | 4 | 10 | 2 |
| Skills, Hungarian | KAPU, collapsed | 15 | 0 | 1 | 0 | 4 | 10 | 7 |
| PLC manuals | KAPU 0.2.1 | 4 | 1 | 1 | 0 | 1 | 1 | 1 |
| PLC manuals | KAPU, collapsed | 3 | 0 | 1 | 0 | 1 | 1 | 21 |

## kapu_ask: what one call opens (J12)

Answered: a frame of the expected page is among the opened ones. Spread: the start of the best door of each of the first 5 rooms, the 2000-token budget shared. Confidence: the best hit alone when it is clearly ahead, otherwise spread, and the reply says confidence: high or low. Tokens without the page: the tokens of the replies that missed it, per question. Search hits: how many fit in the 600-token search reply.

| Question set | Mode | Answered | Reply tokens | Tokens without the page | Search hits |
|---|---|---|---|---|---|
| Rust book | Best hit (collapsed) | 15% | 1,662 | 1,336 | 7.2 |
| Rust book | Best hit (0.2.1) | 15% | 1,653 | 1,315 | 10.0 |
| Rust book | Spread over 5 rooms | 41% | 1,224 | 710 | 7.2 |
| Rust book | Excerpts only, nothing opened | 0% | 215 | 211 | 7.2 |
| Rust book | Confidence, ratio 1.1 | 37% | 1,239 | 835 | 7.2 |
| Rust book | Confidence, ratio 1.3 | 41% | 1,226 | 711 | 7.2 |
| Rust book | Confidence, ratio 1.6 | 41% | 1,226 | 713 | 7.2 |
| Rust book | Best hit, two-line excerpts (J11) | 15% | 1,664 | 1,336 | 5.9 |
| Kubernetes | Best hit (collapsed) | 17% | 1,824 | 1,404 | 8.6 |
| Kubernetes | Best hit (0.2.1) | 17% | 1,800 | 1,391 | 9.9 |
| Kubernetes | Spread over 5 rooms | 39% | 1,130 | 678 | 8.6 |
| Kubernetes | Confidence, ratio 1.3 | 39% | 1,155 | 721 | 8.6 |
| Kubernetes | Best hit, two-line excerpts (J11) | 17% | 1,823 | 1,406 | 7.0 |
| MDN | Best hit (collapsed) | 12% | 1,678 | 1,344 | 9.5 |
| MDN | Best hit (0.2.1) | 11% | 1,614 | 1,331 | 10.0 |
| MDN | Spread over 5 rooms | 22% | 1,124 | 854 | 9.5 |
| MDN | Confidence, ratio 1.3 | 21% | 1,140 | 892 | 9.5 |
| MDN | Best hit, two-line excerpts (J11) | 12% | 1,682 | 1,344 | 7.8 |
| Skills, English | Best hit (collapsed) | 83% | 1,827 | 301 | 5.3 |
| Skills, English | Best hit (0.2.1) | 83% | 1,788 | 295 | 9.8 |
| Skills, English | Spread over 5 rooms | 96% | 1,145 | 51 | 5.3 |
| Skills, English | Excerpts only, nothing opened | 0% | 209 | 210 | 5.3 |
| Skills, English | Confidence, ratio 1.1 | 93% | 1,186 | 85 | 5.3 |
| Skills, English | Confidence, ratio 1.3 | 96% | 1,151 | 52 | 5.3 |
| Skills, English | Confidence, ratio 1.6 | 96% | 1,148 | 51 | 5.3 |

| Question set | Mode | Answered | Reply tokens | Tokens without the page | Search hits |
|---|---|---|---|---|---|
| Skills, English | Best hit, two-line excerpts (J11) | 83% | 1,828 | 300 | 4.7 |
| Skills, translated | Best hit (collapsed) | 76% | 1,794 | 372 | 5.3 |
| Skills, translated | Best hit (0.2.1) | 76% | 1,684 | 384 | 9.6 |
| Skills, translated | Spread over 5 rooms | 81% | 1,144 | 210 | 5.3 |
| Skills, translated | Excerpts only, nothing opened | 0% | 217 | 217 | 5.3 |
| Skills, translated | Confidence, ratio 1.1 | 79% | 1,193 | 249 | 5.3 |
| Skills, translated | Confidence, ratio 1.3 | 81% | 1,148 | 210 | 5.3 |
| Skills, translated | Confidence, ratio 1.6 | 81% | 1,148 | 211 | 5.3 |
| Skills, translated | Best hit, two-line excerpts (J11) | 76% | 1,796 | 372 | 4.6 |

## Do the topic groups match the documentation's folders? (N13)

Second folder level of the source (Kubernetes concepts/workloads, MDN web/css). ARI: 0 is random, 1 the same grouping; NMI in brackets for random groups of the same sizes. k-means: the same number of groups from the room vectors. sim and term: Leiden on only that edge type. Largest group: rooms / tokens.

| Question set | Rooms | Groups / folders | ARI | NMI (random) | k-means ARI | sim only | term only | Largest group |
|---|---|---|---|---|---|---|---|---|
| Kubernetes | 1,722 | 104 / 73 | 0.159 | 0.416 (0.150) | 0.137 | 0.155 | −0.005 | 166 / 1,509,243 |
| MDN | 14,719 | 99 / 50 | 0.040 | 0.390 (0.029) | 0.032 | 0.041 | −0.001 | 733 / 1,296,318 |

- One fix holds and is the default in 0.3.0: one hit per room, with the room's other matching sections named under it. On the test part, all seven sets pooled: room recall@10 +0.022 [95% +0.010, +0.035], MRR@10 +0.011 [+0.008, +0.014], no set worse, search time unchanged (Kubernetes p50 27.3 to 27.9 ms, MDN 313.5 to 313.8 ms).
- Three changes come close but lose one question on a small set, more than the rule allows: separate average lengths for rooms and sections (+0.021 recall@10), BM25 weighted 1.5 with k 20 on top of the collapse (+0.028), and the same with a query-dependent weight (+0.032). They stay switches, off. The graph step, other fusions, stemming variants and title weights give nothing measurable.
- Honest baselines: page-deduplicated chunk BM25 is still ahead of KAPU on Kubernetes (0.484 against 0.451) and MDN (0.322 against 0.288), the chunk hybrid on every set but the PLC manuals. Until 0.2.1 the README compared with chunks counted per chunk. MDN's misses are mostly vocabulary: 112 of 168 pages are in no top 100.

- kapu_ask now opens the start of the first 5 rooms in one shared budget: the expected room is in the reply for 42% instead of 15% (Rust book), 39% instead of 17% (Kubernetes), 22% instead of 12% (MDN) and 96% instead of 83% (skills), in about 1,130 instead of 1,800 tokens. The planned confidence signal is not calibrated (where it says high, 11% on Kubernetes and MDN), so it is not shown.
- On PLC manuals a model does not know (six vendors, 623 pages), Claude Sonnet answered 92% correctly with the KAPU tools, 97% with the right page in the prompt and 28% alone. For 24 questions the manuals do not answer, the KAPU agent said so 23 times; alone the model made an answer up 11 times. One run per question; the question writer saw the pages.
- The graph helps navigation, not ranking: of 27 later opens in the logged live runs, 5 (19%) came from a neighbour list (link 2 of 64 shown, sim 3 of 64, term 0 of 221), at 70 tokens per opened door. The topic groups follow the documentation's folders at the level of k-means on the room vectors, and the sim edges carry that structure.

## File format 1.2: vectors in the file, replies within a budget

Measured 2026-10-26 with tools 0.2.0, specification v1.2, file format 1.2, on the same corpora, questions and machine as format 1.1, and on a fourth corpus: agent skills and frontend notes with 77 English and 60 Hungarian questions. The search vectors are now in the file, one sign bit per component of every window vector. The sign bits pick the 200 best candidates and the model's float vectors order them, so the first search on another machine needs no cache. Every MCP reply starts with its own price and stays within a max_tokens budget, and kapu_ask searches and opens in one call.

## Requirements of the second protocol

The measurable requirements of the 2026-10-26 protocol and their acceptance limits. Tokens: o200k_base.

| ID | Requirement | Target | Measured | Met |
|---|---|---|---|---|
| K-01 | Tool descriptions and input schemas, all six tools | ≤ 800 | 797 | yes |
| K-02 | First map reply on a file of 3,000 rooms | ≤ 1,000 | 306 | yes |
| K-05 | Search reply, 10 hits with excerpts (largest on any corpus) | ≤ 600 | 586 | yes |
| K-07 | Questions answered by one kapu_ask call (skills corpus) | ≥ 70% | 83.1% | yes |
| K-11 | File growth from the #VEC section (largest corpus, MDN) | ≤ 3% | 3.3% | no |
| K-12 | Hungarian questions on English skills, translated: hybrid recall@10 | ≥ 0.75 | 0.78 | yes |
| K-13 | The same without translation (needs another embedding model) | ≥ 0.60 | 0.55 | no |
| K-22 | Opening a door on a file of 3,000 rooms (median) | < 20 ms | 7.6 ms | yes |

## Four corpora in format 1.2

File size before and after the vector section. Times and reply sizes are medians over MCP, except the largest search reply; tokens are what the agent receives.

| Corpus | File, v1.1 (MiB) | File, v1.2 (MiB) | Growth | Search (ms) | Open (ms) | First map (tokens) | Next map (tokens) | Search (tokens) | Largest search (tokens) | kapu_ask (tokens) | Answered by one kapu_ask |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Rust book | 2.21 | 2.24 | 1.7% | 32 | 3.6 | 328 | 159 | 527 | 580 | 1,657 | 21% |
| Kubernetes | 26.22 | 26.78 | 2.2% | 50 | 7.9 | 967 | 983 | 528 | 586 | 1,795 | 20% |
| MDN | 128.34 | 132.52 | 3.3% | 282 | 48 | 969 | 980 | 500 | 583 | 1,664 | 12% |
| Skills and notes | 5.56 | 5.67 | 2.0% | 26 | 2.9 | 600 | 224 | 547 | 585 | 1,786 | 83% |

### Retrieval: room-level recall@10, hybrid ranking

Sign bits alone lost on MDN, on the skills and most of all with Hungarian questions on English text. With the float rescoring of the best candidates they are at the float level or above everywhere; the price is search time.

| Question set | Questions | v1.1, float vectors | v1.2, sign bits alone | v1.2, sign bits + rescoring |
|---|---|---|---|---|
| Rust book (Stack Overflow) | 68 | 0.485 | 0.500 | 0.485 |
| Kubernetes (Stack Overflow) | 201 | 0.458 | 0.458 | 0.458 |
| MDN (Stack Overflow) | 395 | 0.271 | 0.246 | **0.278** |
| Skills, English questions | 77 | 0.948 | 0.922 | **0.961** |
| Skills, Hungarian questions | 60 | 0.633 | 0.583 | 0.633 |
| Skills, the Hungarian questions translated | 60 | — | 0.833 | 0.833 |

### Not measured yet

- Skill finding with 250 skills (the test machine has 91), and with a prompt that does not mention skills, where the agent has to decide on its own.
- A local model (Ollama) as the agent.
- Clients other than Claude Code: the configurations for the Codex CLI and VS Code are in the README, but they were not tried, and no measurement with them is planned.

## Finding skills: native Claude Code skills, one KAPU file, a Markdown folder

12 realistic development tasks (a landing page, an MCP server, a Claude Code plugin, a flaky API, Office files and others), each with 2 to 4 skills fixed in advance as clearly relevant, 39 in all. The same skills in three forms, 45 and 91 of them (91 is every real skill on the test machine). Every task ran once per form and size with opus through claude -p (Claude Code 2.1.310), on 2026-10-26. The prompt points at the skills, as a user would: load every relevant skill with the Skill tool; find and open every relevant skill in the KAPU file; find and read every relevant skill in this folder. A skill counts as found when the Skill tool loaded it, one of its files was read, or one of its rooms was opened.

## Results per form

Wrong: loaded although the task neither needs it nor could reasonably use it. Tokens per task: every input token of the run, each turn counted (median). $ per task: what Claude Code reports at API prices, cache reads at their lower price (median). Fixed cost: the first turn of a session that needs no skill; the native skill list is in every session's system prompt.

| Skills | Form | Relevant skills found | Tasks with all found | Wrong skills loaded | Tokens per task | $ per task | Calls per task | Seconds per task | Fixed cost per session (tokens) |
|---|---|---|---|---|---|---|---|---|---|
| 45 | Native skills | 38 / 39 (97%) | 11 / 12 | 0 | 42,574 | $0.224 | 4 | 9 | 12,652 |
| 45 | KAPU file, catalog (0.2.1) | 38 / 39 (97%) | 11 / 12 | 3 | 58,266 | $0.131 | 5.5 | 25 | 4,775 |
| 45 | KAPU file (0.2.0) | 37 / 39 (95%) | 10 / 12 | 0 | 77,215 | $0.229 | 7.5 | 30 | 4,774 |
| 45 | Markdown folder | 39 / 39 (100%) | 12 / 12 | 14 | 93,176 | $0.283 | 10.5 | 24 | 5,430 |
| 91 | Native skills | 39 / 39 (100%) | 12 / 12 | 0 | 44,953 | $0.236 | 4 | 12 | 14,081 |
| 91 | KAPU file, catalog (0.2.1) | 38 / 39 (97%) | 11 / 12 | 3 | 48,972 | $0.099 | 5 | 22 | 4,775 |
| 91 | KAPU file (0.2.0) | 38 / 39 (97%) | 11 / 12 | 0 | 88,418 | $0.255 | 9 | 33 | 4,774 |
| 91 | Markdown folder | 39 / 39 (100%) | 12 / 12 | 12 | 107,726 | $0.336 | 10.5 | 33 | 5,430 |

- With a catalog of the skills in one reply (tools 0.2.1: kapu_map kind=skill, D154), the KAPU file finds as many as before (38 of 39 with 91 skills) in 5 calls instead of 9, with 48,972 input tokens per task (before: 88,418; native skills: 44,953). At $0.099 per task it costs less than the native skills ($0.236), for two reasons: it opens the first part of each skill (up to 2,000 tokens; the rest on request) where the Skill tool loads the whole skill, and less new text is written to the prompt cache, which costs twice the input price, while cache reads cost a tenth.
- The native skills still find the most (39 of 39 with 91 skills) and load nothing wrong. The KAPU file loaded 3 skills that one task did not need, and missed one skill (frontend-expert for the Figma task, which the native skills also missed with 45 skills). The Hungarian-language skill it missed before, it now finds from the catalog.
- The Markdown folder found every skill, but also read 12 skills the tasks did not need (with 91 skills) and cost the most ($0.336 per task).
- The fixed cost of a session: 4,775 tokens with the KAPU tools whatever the number of skills, against 12,652 with 45 and 14,081 with 91 native skills. 250 skills were not measured: the test machine has only 91. Only the KAPU runs were repeated; the native and Markdown runs are those of the first round.

# File format 1.1: smaller files, faster calls, better search by meaning

Three changes after the pilot, measured again on the same corpora and questions. The index stores a door's position instead of its id, so files are about 4.5 times smaller and the 13-million-token MDN corpus now builds with the released tool. One call parses and checks the file once and finds index lines by binary search. Search by meaning uses one vector per 512-token window of a door, and the best window counts.

## Before and after

Room recall@10: any part of the expected page among the first 10 hits. MDN with v1.0.4 was an experimental build with a 64 MiB line limit.

| Metric | Corpus | v1.0.4 | v1.1 |
|---|---|---|---|
| File size (MiB) | Rust book | 7.6 | 2.2 |
| File size (MiB) | Kubernetes | 117.4 | 26.2 |
| File size (MiB) | MDN | 542.2 | 128.3 |
| Builds with the released tool | MDN | no (E005) | yes |
| Search over MCP (ms, median) | Rust book | 56 | 12.8 |
| Search over MCP (ms, median) | Kubernetes | 657 | 19.2 |
| Search over MCP (ms, median) | MDN | 3,172 | 158.2 |
| Opening a door over MCP (ms, median) | Rust book | 17 | 2.4 |
| Opening a door over MCP (ms, median) | Kubernetes | 216 | 4.9 |
| Opening a door over MCP (ms, median) | MDN | 1,040 | 44.1 |
| Room recall@10, hybrid search | Rust book | 0.5 | 0.5 |
| Room recall@10, hybrid search | Kubernetes | 0.4 | 0.5 |
| Room recall@10, hybrid search | MDN | 0.3 | 0.3 |
| Room recall@10, vectors only | Rust book | 0.3 | 0.4 |
| Room recall@10, vectors only | Kubernetes | 0.3 | 0.3 |
| Room recall@10, vectors only | MDN | 0.2 | 0.1 |
| kapu bench run time (s) | Kubernetes | 187 | 8 |
| kapu bench run time (s) | MDN | 1,774 | 127 |

# Same content in three formats: KAPU, one Markdown file, a PDF

The PixelLab documentation (third-party, not published), 12 requests of a beginner indie developer. The Markdown and PDF agents searched the text and read only what they found: lines of the Markdown file, pages of the PDF. Pasting the whole file into the conversation would cost 206,899 tokens on every request. Three blind Claude Sonnet judges checked every claim against the documentation.

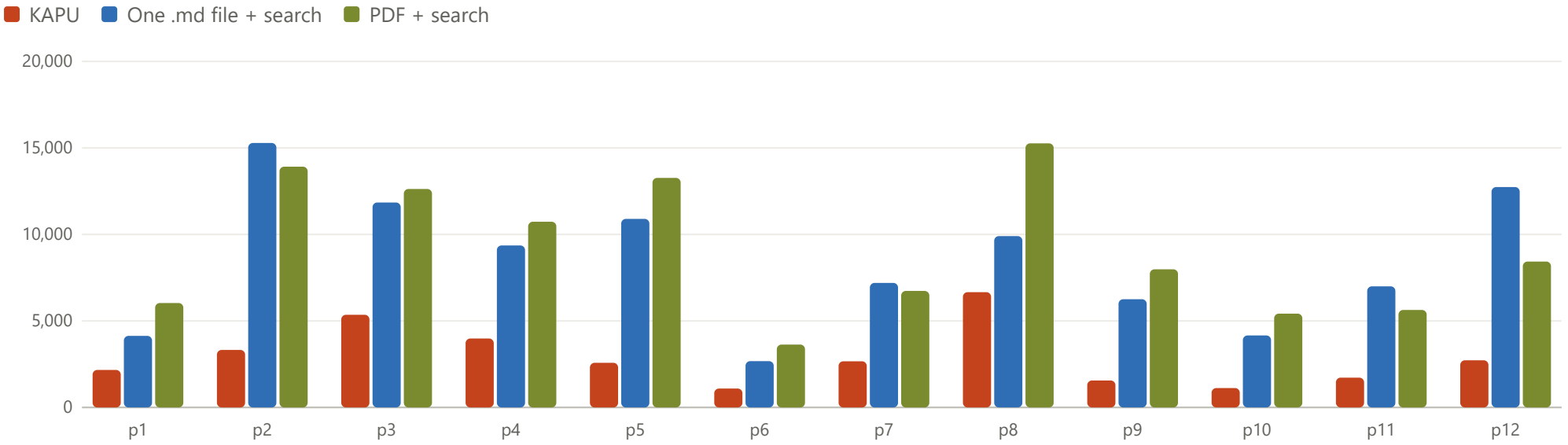
## Second round: the same prompt for all three formats

KAPU with specification v1.0.4. The Markdown and PDF answers are those of the first round, scored again by new judges.

| Format | Correct | Partly | Supported claims | All claims | Helpfulness (1-5) | Tokens/request | Tokens, 12 requests | Calls/request |
|---|---|---|---|---|---|---|---|---|
| KAPU | 12 | 0 | 153 | 157 | 4.17 | 4,104 | 48,557 | 4 |
| One .md file + search | 12 | 0 | 140 | 145 | 4.00 | 8,280 | 101,452 | 8 |
| PDF + search | 11 | 1 | 147 | 151 | 3.83 | 8,207 | 109,675 | 6 |

### First round: tokens read per request

KAPU with specification v1.0.3 and its own prompt. Tokens: what the tools returned to the agent (o200k_base).



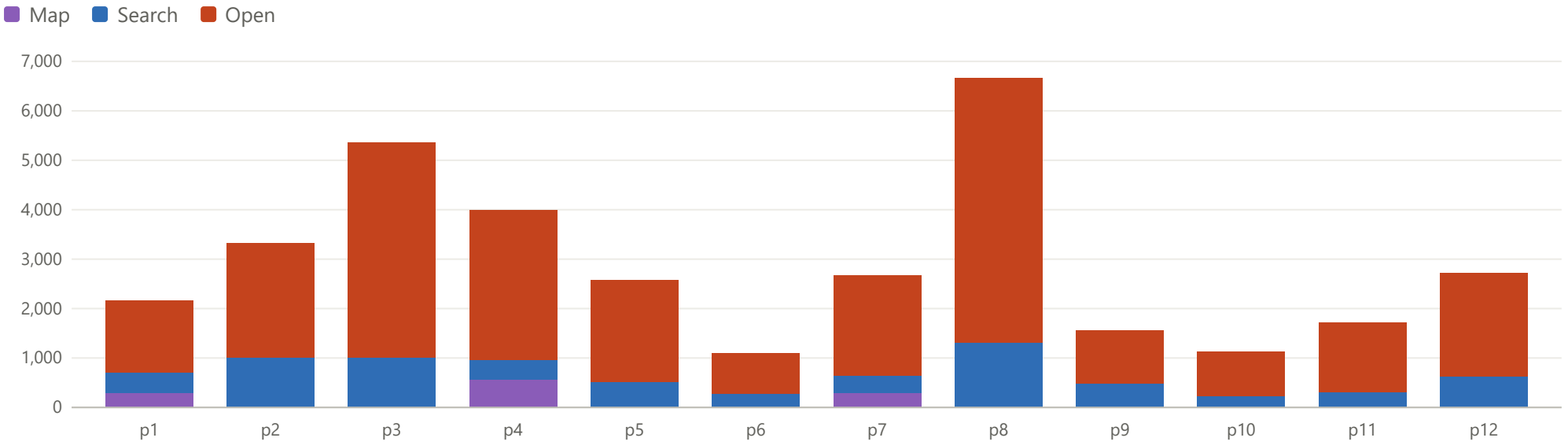

| Format | Correct | Partly | Supported claims | All claims | Helpfulness (1-5) | Tokens/request | Tokens, 12 requests | Calls/request |
|---|---|---|---|---|---|---|---|---|
| KAPU | 12 | 0 | 122 | 123 | 4.08 | 2,624 | 34,953 | 3 |
| One .md file + search | 12 | 0 | 134 | 138 | 4.25 | 8,280 | 101,452 | 8 |
| PDF + search | 11 | 1 | 146 | 151 | 4.33 | 8,207 | 109,675 | 6 |

## Why thousands of tokens when the map is a few hundred?

The map and the search are the table of contents and the hit list. Most tokens come from the opened doors: the documentation text the answer is written from. Of the KAPU tokens of the 12 requests, opening doors took 77%, searching 20%, the map 3%.

### KAPU tokens per request, by call

A typical request: one or two searches (300-400 tokens), one or two opened doors (1,000-3,000 tokens), rarely the map.




## Scale: small, medium and large documentation

S: the Rust book, M: the Kubernetes documentation, L: MDN Web Docs, all in format 1.1. Over one MCP session: the map, every group list, 20 searches with room titles as queries and an open of each top hit.

### Build, size and responses

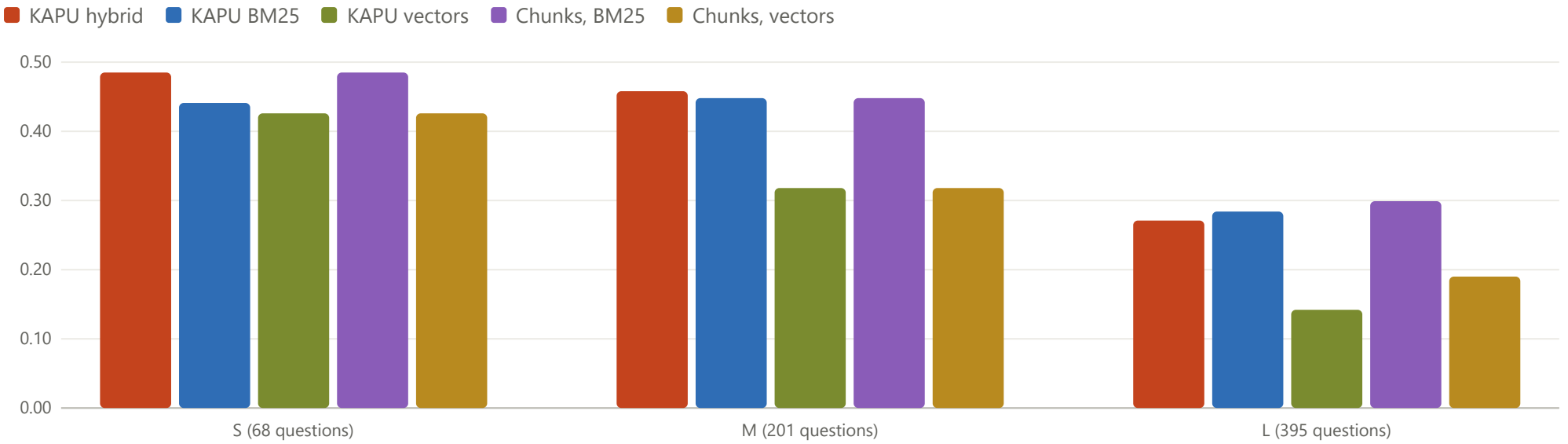
| Size | Corpus | Knowledge (tokens) | Rooms | Doors | Groups | Build (s) | Memory (GB) | File (MiB) | Map (tokens) | First map (tokens) | Largest group list (tokens) | Search (tokens) | Search (ms) | First search (s) | Open (tokens) | Open (ms) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| S | Rust book | 321,618 | 111 | 231 | 7 | 4.6 | 0.69 | 2.2 | 153 | 322 | 804 | 246 | 12.8 | 1.1 | 1,689 | 2.4 |
| M | Kubernetes | 3,873,333 | 1,722 | 6,370 | 104 | 38 | 1.03 | 26.2 | 1,950 | 2,318 | 6,957 | 260 | 19.2 | 1 | 240 | 4.9 |
| L | MDN | 12,995,136 | 14,719 | 84,910 | 99 | 240.3 | 2.68 | 128.3 | 1,921 | 2,303 | 30,118 | 212 | 158.2 | 1.3 | 366 | 44.1 |

## Retrieval without an agent: KAPU against chunked RAG

Real Stack Overflow questions whose accepted answer links exactly one page of the documentation: 68, 201 and 395 questions. A hit is any part of that page among the first results. The chunk baselines use 512-token chunks with 64 tokens of overlap and the same embedding model.

## Room recall@10 by size and method

KAPU returns doors (rooms and their sections), the baselines return chunks.




| Size | Method | Questions | Recall@5 | Recall@10 | MRR@10 |
|---|---|---|---|---|---|
| S | KAPU hybrid | 68 | 0.397 | 0.485 | 0.281 |
| S | KAPU BM25 | 68 | 0.338 | 0.441 | 0.259 |
| S | KAPU vectors | 68 | 0.353 | 0.426 | 0.184 |
| S | Chunks, BM25 | 68 | 0.426 | 0.485 | 0.282 |
| S | Chunks, vectors | 68 | 0.368 | 0.426 | 0.230 |
| M | KAPU hybrid | 201 | 0.373 | 0.458 | 0.245 |
| M | KAPU BM25 | 201 | 0.348 | 0.448 | 0.218 |
| M | KAPU vectors | 201 | 0.219 | 0.318 | 0.147 |
| M | Chunks, BM25 | 201 | 0.338 | 0.448 | 0.218 |
| M | Chunks, vectors | 201 | 0.249 | 0.318 | 0.145 |
| L | KAPU hybrid | 395 | 0.172 | 0.271 | 0.128 |
| L | KAPU BM25 | 395 | 0.220 | 0.284 | 0.135 |
| L | KAPU vectors | 395 | 0.086 | 0.142 | 0.055 |
| L | Chunks, BM25 | 395 | 0.230 | 0.299 | 0.159 |

| Size | Method | Questions | Recall@5 | Recall@10 | MRR@10 |
|---|---|---|---|---|---|
| L | Chunks, vectors | 395 | 0.124 | 0.190 | 0.081 |

## Live run on the Rust book: five systems, 30 questions

The model answered these popular questions well from memory too (closed book: 30 of 30), so here accuracy does not separate the systems; the cost does. KAPU with specification v1.0.3; with v1.0.4 the KAPU agents read a median of 2,331 tokens per question at the same accuracy.

### Correct answers and cost per system

Two blind judges (Claude Sonnet and Haiku). All agent tokens: the whole run divided by its questions, an estimate.

| System | Correct (Sonnet judge) | Correct (Haiku judge) | Questions | Tokens read/question | Calls/question | All agent tokens/question |
|---|---|---|---|---|---|---|
| Closed book, no tools | 30 | 28 | 30 | 0 | 0 | 2,706 |
| Agentic chunked RAG | 30 | 27 | 30 | 4,150 | 1 | 12,648 |
| grep/read on the Markdown folder | 29 | 28 | 30 | 1,290 | 2 | 9,343 |
| llms.txt | 30 | 29 | 30 | 8,738 | 2 | 11,615 |
| KAPU | 30 | 29 | 30 | 3,982 | 2 | 10,118 |

## Earlier fix round: specification v1.0.4

| Metric | v1.0.3 | v1.0.4 |
|---|---|---|
| KAPU on the Rust book: tokens read per question (median) | 3,982 | 2,331 |
| KAPU on the Rust book: correct answers (of 30) | 30 | 30 |
| First map call, Kubernetes (tokens) | 4,989 | 2,318 |
| First map call, MDN (tokens) | 4,139 | 2,303 |

## Method and limits

- Corpora: the Rust book (MIT or Apache-2.0), the Kubernetes documentation (CC BY 4.0) and MDN Web Docs (CC BY-SA 2.5) at pinned commits. Questions: Stack Overflow API (CC BY-SA); only their ids are stored.
- Agents: Claude Sonnet subagents that could only use logged tools. Tokens are counted with o200k_base.
- Judges: blind Claude models; the answers were shuffled under random letters, and source lines were removed.
- This is a pilot on one machine (Windows 11, release build), not a benchmark. The PixelLab file is third-party and is not published.

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Repository RESULTS.md Raw numbers (JSON)